Z- TRANSFORMS & DIFFERENCE EQUATIONS

$(1-x)^{-1} = 1+x+x^2+x^3+x^4+x^5+ \dots$

$(1+x)^{-1} = 1-x+x^2-x^3+x^4-x^5+ \dots$

$(1-x)^{-2} = 1+2x+3x^2+4x^3+5x^4+ \dots$

$(1+x)^{-2} = 1-2x+3x^2-4x^3+5x^4- \dots$

If $U_n = f(n)$ be function in variable 'n' defined for $n = 0, 1, 2, 3 \dots$ and $U_n = 0$ for $n<0$ then the Z-transform of U_n is denoted by $Z_T(U_n)$ and it is given by

$$\boxed{Z_T(U_n) = \sum_{n=0}^{\infty} U_n Z^{-n}}$$

Note:- $Z_T(n^k) = -z \cdot \frac{d}{dz}\left[Z_T(n^{k-1})\right]$

Z_T of some standard functions

1] $Z_T(k^n)$

we have

$$Z_T(U_n) = \sum_{n=0}^{\infty} U_n Z^{-n}$$

from,

$$Z_T(k^n) = \sum_{n=0}^{\infty} k^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{k}{z}\right)^n$$

$$= \left(\frac{k}{z}\right)^0 + \left(\frac{k}{z}\right)^1 + \left(\frac{k}{z}\right)^2 + \left(\frac{k}{z}\right)^3 + \dots$$

$$= 1 + \left(\frac{k}{z}\right) + \left(\frac{k}{z}\right)^2 + \left(\frac{k}{z}\right)^3 + \dots$$

$$= \left(1-\frac{k}{z}\right)^{-1} \quad (\because (1-x)^{-1} = 1+x+x^2+x^3+\dots)$$

$$= \frac{1}{\left(1-\frac{k}{z}\right)} \Rightarrow \frac{1}{\frac{z-k}{z}}$$

$$\boxed{Z_T(k^n) = \frac{z}{z-k}}$$

2. $Z_T(1)$

we have $Z_T(k^n) = \dfrac{z}{z-k}$

put $k=1$

$$Z_T(1^n) = \frac{z}{z-1}$$

$$\boxed{Z_T(1) = \frac{z}{z-1}}$$

3. $Z_T(n)$

we have

$$Z_T(n^k) = -z\frac{d}{dz}\left(Z_T(n^{k-1})\right)$$

put $k=1$

$$Z_T(n^1) = -z\frac{d}{dz}\left(Z_T(n^{1-1})\right)$$

$$= -z\frac{d}{dz}\left[Z_T(1)\right]$$

$$= -z\frac{d}{dz}\left[\frac{z}{z-1}\right]$$

$$= -z\left[\frac{1}{(z-1)^2}(z-1)(1) - z(1)\right] \Rightarrow \left[\because \frac{du}{dv} = \frac{1}{v^2}\left[vdu - udv\right]\right]$$

$$= \frac{-z(z-1-z)}{(z-1)^2}$$

$$\boxed{Z_T(n) = \frac{z}{(z-1)^2}}$$

4. $Z_T(n^2)$

we have. $Z_T(n^k) = -z\dfrac{d}{dz}\left[Z_T(n^{k-1})\right]$

put $k=2$.

$$Z_T(n^2) = -z\frac{d}{dz}\left[Z_T(n^{2-1})\right]$$

$$= -z\frac{d}{dz}\left[Z_T(n)\right] = -z\frac{d}{dz}\left[\frac{z}{(z-1)^2}\right]$$

$$= -z \frac{1}{((z-1)^2)^2}\left[(z-1)^2\cdot(1) - z\,2(z-1)\right] \qquad d\left(\frac{u}{v}\right) = \frac{1}{v^2}[v\,du - u\,dv]$$

$$= \frac{-z\,\cancel{(z-1)}\left[(z-1) - 2z\right]}{(z-1)^{\cancel{4}}} = \frac{-z(z-1-2z)}{(z-1)^3}$$

$$= \frac{-z(-z-1)}{(z-1)^3}$$

$$\boxed{Z_T(n^2) = \frac{z^2+z}{(z-1)^3}}$$

5. $Z_T(n^3)$

we have $Z_T(n^k) = -z\frac{d}{dz}\left[Z_T(n^{k-1})\right]$

put $k=3$

$$Z_T(n^3) = -z\frac{d}{dz}\left[Z_T(n^{3-2})\right]$$

$$= -z\frac{d}{dz}\left[Z_T(n^2)\right] \Rightarrow -z\frac{d}{dz}\left[\frac{z^2+z}{(z-1)^3}\right]$$

$$= -z\frac{1}{((z-1)^3)^2}\left[(z-1)^3(2z+1) - (z^2+z)3(z-1)^2\right] \qquad d\left(\frac{u}{v}\right) = \frac{1}{v^2}\,v\,du - u\,dv$$

$$= \frac{-z\,\cancel{(z-1)^2}\left[(z-1)(2z+1) - 3(z^2+z)\right]}{\cancel{(z-1)^6}}$$

$$= \frac{-z\left[2z^2+z-2z-1-3z^2-3z\right]}{(z-1)^4}$$

$$= \frac{-z(-z^2-4z-1)}{(z-1)^4}$$

$$\boxed{Z_T(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}}$$

Damping property

If z-transform $Z_T(U_n) = F(z)$ then

i) $Z_T(k^n U_n) = F\left(\frac{z}{k}\right)$

ii) $Z_T(k^{-n} U_n) = F(kz)$

Note:- $Z_T(k^n \cdot n) = \frac{kz}{(z-k)^2}$

$$Z_T(k^n \cdot n^2) = \frac{kz^2 + k^2 z}{(z-k)^3}$$

$$Z_T(k^n\, n^3) = \frac{kz^3 + 4k^2z^2 + k^3 z}{(z-k)^4}$$

List of formulae:

1. $Z_T(k^n) = \frac{z}{z-k}$

2. $Z_T(1) = \frac{z}{z-1}$

3. $Z_T(n) = \frac{z}{(z-1)^2}$

4. $Z_T(n^2) = \frac{z^2+z}{(z-1)^3}$

5. $Z_T(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$

6. $Z_T(\sin(n\theta)) = \frac{z\sin\theta}{z^2 2z\cos\theta+1}$

7. $Z_T[\cos(n\theta)] = \frac{z^2 - z\cos\theta}{z^2 2z\cos\theta+1}$

8. $Z_T[\sinh(n\theta)] = \frac{z\sinh\theta}{z^2 2z\cosh\theta+1}$

9. $Z_T[\cosh(n\theta)] = \frac{z^2 - z\cosh\theta}{z^2 2z\cosh\theta+1}$

v. ~~Obtain~~ $e^{in\theta} = \cos(n\theta) + i\sin(n\theta)$

$$e^{-in\theta} = \cos(n\theta) - i\sin(n\theta)$$

$$\sin(n\theta) = \frac{e^{in\theta} - e^{-in\theta}}{2i}$$

$$\cos n\theta = \frac{e^{in\theta} + e^{-in\theta}}{2}$$

$$\cosh n\theta = \frac{e^{n\theta} + e^{-n\theta}}{2}$$

$$\sinh n\theta = \frac{e^{n\theta} - e^{-n\theta}}{2}.$$

*** 1. Obtain z-transform of $\cos n\theta$ & $\sin n\theta$

$\Rightarrow$ Let $e^{in\theta} = \cos n\theta + i\sin n\theta$

we have $Z_T(k^n) = \frac{z}{z-k}$

Now, $Z_T(e^{in\theta}) = Z_T\left[(e^{i\theta})^n\right]$

$$= \frac{z}{z-e^{i\theta}}$$

$$= \frac{z}{z-e^{i\theta}} \times \frac{z-e^{-i\theta}}{z-e^{-i\theta}}$$

$$= \frac{z^2 - z(e^{-i\theta})}{z^2 - ze^{-i\theta} - ze^{i\theta} + e^{i\theta} \cdot e^{-i\theta}}$$

$e^{i\theta} \cdot e^{-i\theta}$

$e^{i\theta - i\theta}$

$e^0 = 1$

$$Z_T\left[e^{in\theta}\right] = \frac{z^2 - z(\cos\theta - i\sin\theta)}{z^2 - z(e^{i\theta} + e^{-i\theta}) + 1}$$

$$Z_T\left[\cos(n\theta) + i\sin(n\theta)\right] = \frac{z^2 - z\cos\theta + iz\sin\theta}{z^2 - 2z\cos\theta + 1}$$

$$Z_T\left[\cos(n\theta)\right] + i\, Z_T\left[\sin(n\theta)\right] = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1} + i\,\frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$$

On comparing real & imaginary parts on b.s

$$Z_T[\cos(n\theta)] = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1} \; ; \; Z_T[\sin(n\theta)] = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Q2. Obtain the z-transform of $\cosh(n\theta)$ & $\sinh(n\theta)$

$\Rightarrow$ we have $\cosh n\theta = \dfrac{e^{n\theta} + e^{-n\theta}}{2}$

$$Z_T[\cosh(n\theta)] = Z_T\left[\frac{e^{n\theta} + e^{-n\theta}}{2}\right]$$

$$= \frac{1}{2} Z_T\left[e^{n\theta} + e^{-n\theta}\right]$$

$$= \frac{1}{2}\left[Z_T (e^{\theta})^n + Z_T (e^{-\theta})^n\right]$$

$$= \frac{1}{2}\left[\frac{z}{z - e^{\theta}} + \frac{z}{z - e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{1}{z - e^{\theta}} + \frac{1}{z - e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{z - e^{-\theta} + (z - e^{\theta})}{(z - e^{\theta})(z - e^{-\theta})}\right]$$

$e^{i\theta} \cdot e^{-i\theta}$

$e^{i\theta - i\theta}$

$e^{0} = 1$

$$= \frac{z}{2}\left[\frac{2z - (e^{\theta} + e^{-\theta})}{z^2 - ze^{-\theta} - ze^{\theta} + e^{\theta} \cdot e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{2z - (2\cosh\theta)}{z^2 - z(e^{\theta} + e^{-\theta}) + 1}\right]$$

$$= \frac{z}{\cancel{2}}\left[\frac{\cancel{2}(z - \cosh\theta)}{z^2 - z(2\cosh\theta) + 1}\right]$$

$$\boxed{Z_T[\cosh(n\theta)] = \frac{z^2 - z\cosh\theta}{z^2 - 2z\cosh\theta + 1}}$$

$$Z_T[\sinh(n\theta)] = \frac{e^{n\theta} - e^{-n\theta}}{2}$$

$$Z_T[\sinh(n\theta)] = Z_T\left[\frac{e^{n\theta} - e^{-n\theta}}{2}\right]$$

$$= \frac{1}{2} Z_T\left[e^{n\theta} - e^{-n\theta}\right]$$

$$= \frac{1}{2}\left[Z_T[(e^{\theta})^n] - Z_T(e^{-\theta})^n\right]$$

$$= \frac{1}{2}\left[\frac{z}{z-e^{\theta}} - \frac{z}{z-e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{1}{z-e^{\theta}} - \frac{1}{z-e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{z-e^{-\theta}-(z-e^{\theta})}{(z-e^{\theta})(z-e^{-\theta})}\right]$$

$$= \frac{z}{2}\left[\frac{z-e^{-\theta}-z+e^{\theta}}{z^2\, ze^{-\theta}-ze^{\theta}+e^{\theta}.e^{-\theta}}\right]$$

$$= \frac{z}{2}\left[\frac{e^{\theta}-e^{-\theta}}{z^2-z\,(e^{\theta}+e^{-\theta})+1}\right]$$

$$= \frac{z}{2}\;\frac{2\sinh(\theta)}{z^2 z\, 2\cosh\theta+1}$$

$$\boxed{Z_T[\sinh(n\theta)] = \frac{z\sinh\theta}{z^2\; 2z\cosh\theta+1}}$$

03. Find the z-transform of $(n+1)^2$

$\rightarrow$

$$(n+1)^2 = n^2+1+2n$$

apply z transform on b.s

$$Z_T[(n+1)^2] = Z_T[n^2+1+2n]$$

$$= Z_T[n^2] + Z_T(1) + 2Z_1(n)$$

$$= \frac{z^2+z}{(z+1)^2} + \frac{z}{z-1} + 2\left[\frac{z}{(z-1)^2}\right]$$

$$= \frac{z^2+z+z(-1)^2+2z(z-1)}{(z-1)^3}$$

$$= \frac{z^2+z+z(z^2+1-2z)+2z^2-2z}{(z-1)^3}$$

$$= \frac{z^2+\cancel{z}+z^3+\cancel{z}-\cancel{2z^2}+\cancel{2z^2}-\cancel{2z}}{(z-1)^3} = \frac{z^3+z^2}{(z-1)^3}$$

04. Find the Z-transform of $\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)$

=> WKT

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

Now, $\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right) = \cos\left(\frac{n\pi}{2}\right)\cdot\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{n\pi}{2}\right)\sin\left(\frac{\pi}{4}\right)$

$\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{\sqrt{2}}\sin\left(\frac{n\pi}{2}\right)$

apply z-transform on b.s

$$Z_T\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right) = Z_T\left[\frac{1}{\sqrt{2}}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{\sqrt{2}}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{1}{\sqrt{2}} Z_T\left[\cos\left(\frac{n\pi}{2}\right)\right] - \frac{1}{\sqrt{2}} Z_T\left[\sin\left(\frac{n\pi}{2}\right)\right]$$

we have, $Z_T[\cos(n\theta)] = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1}$

$$Z_T\left[\cos\left(\frac{n\pi}{2}\right)\right] = \frac{z^2 - z\cos(\pi/2)}{z^2 - 2z\cos(\pi/2) + 1}$$

$$Z_T\left[\cos\left(\frac{n\pi}{2}\right)\right] = \frac{z^2 - z(0)}{z^2 - 2z\times 0 + 1} = \frac{z^2}{z^2+1}$$

again, $Z_T[\sin(n\theta)] = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$

$Z_T(\cos n\theta) = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1}$

$$Z_T\left[\sin\left(\frac{n\pi}{2}\right)\right] = \frac{z\sin(\pi/2)}{z^2 - 2z\cos(\pi/2) + 1}$$

$Z_T(\sin n\theta) = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$

$$= \frac{z(1)}{z^2 - 2z(0) + 1} = \frac{z}{z^2+1}$$

$$Z_T\left[\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)\right] = \frac{1}{\sqrt{2}}\frac{z^2}{z^2+1} - \frac{1}{\sqrt{2}}\frac{z}{z^2+1}$$

$$= \frac{z^2 - z}{\sqrt{2}(z^2+1)}$$

5. Find the Z-transform of $\sin(3n+5)$

$\Rightarrow$ we have

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\sin(3n+5) = \sin(3n).\cos(5) + \cos(3n).\sin(5)$

taking z-transform on b.s

$$Z_T[\sin(3n+5)] = Z_T[\sin(3n).\cos(5) + \cos(3n).\sin(5)]$$

$$= \cos(5)\, Z_T[\sin(3n)] + \sin(5).\, Z_T[\cos(3n)] \rightarrow ①$$

we have, $Z_T[\sin n\theta] = \dfrac{z\sin\theta}{z^2\, 2z\cos\theta+1}$

$$\boxed{Z_T[\sin(3n)] = \frac{z\sin(3)}{z^2 - 2z\cos(3)+1}}$$

also, $Z_T[\cos(n\theta)] = \dfrac{z^2 - z\cos\theta}{z^2\, 2z\cos\theta+1}$

$$\boxed{Z_T[\cos(3n)] = \frac{z^2 - z\cos(3n)}{z^2\, 2z\cos(3)+1}}$$

eq^n ① $\Rightarrow$

$$Z_T[\sin(3n+5)] = \frac{\cos(5).\, z\sin(3)}{z^2\, 2z\cos 3+1} + \frac{\sin(5)\, z^2 - z\cos(3)}{z^2\, 2z\cos(3)+1}$$

$$= \frac{z\sin(3).\cos(5) + z^2\sin(5) - z\cos(3)\sin(5)}{z^2 - 2z\cos(3)+1}$$

$$= \frac{z[\sin(3)\cos(5) - \cos(3).\sin(5)] + z^2\sin(5)}{z^2\, 2z\cos(3)+1}$$

$$= \frac{z[\sin(3-5)] + z^2\sin(5)}{z^2\, 2z(\cos(3))+1}$$

$$= \frac{z(\sin(-2)) + z^2\sin 5}{z^2\, 2z\cos(3)+1}$$

$$= \frac{z^2\sin 5 - z\sin(2)}{z^2 - 2z\cos(3)+1}$$

6. Find z-transform of $2n + \sin\left(\frac{n\pi}{4}\right) + 1$

$\Rightarrow \quad Z_T\left[2n + \sin\left(\frac{n\pi}{4}\right) + 1\right]$

$$= 2Z_T(n) + Z_T\left[\sin\left(\frac{n\pi}{4}\right)\right] + Z_T(1)$$

$$= 2\left[\frac{z}{(z-1)^2}\right] + Z_T\left[\sin\left(\frac{n\pi}{4}\right)\right] + \frac{z}{z-1} \rightarrow (1)$$

we have, $Z_T\left[\sin(n\theta)\right] = \frac{z\sin\theta}{z^2 2z\cos\theta + 1}$

$$Z_T\left[\sin\left(\frac{n\pi}{4}\right)\right] = \frac{z.\sin(\pi/4)}{z^2 2z\cos\left(\frac{\pi}{4}\right) + 1}$$

$$= \frac{z\frac{1}{\sqrt{2}}}{z^2 2z\left(\frac{1}{\sqrt{2}}\right) + 1}$$

$$= \frac{\frac{z}{\sqrt{2}}}{\frac{\sqrt{2}z^2 - 2z + \sqrt{2}}{\sqrt{2}}}$$

$$Z_T\left[\sin\left(\frac{n\pi}{4}\right)\right] = \frac{z}{\sqrt{2}z^2 - 2z + \sqrt{2}}$$

eqn (1) $\Rightarrow Z_T\left[2n + \sin\left(\frac{n\pi}{4}\right) + 1\right] = \frac{2z}{(z-1)^2} + \frac{z}{\sqrt{2}z^2 - 2z + \sqrt{2}} + \frac{z}{z-1}$

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7. Find z-transform of $5n^2 + 4\cos\left(\frac{n\pi}{2}\right) - 4^{n+2}$.

$\Rightarrow \quad 5n^2 + 4\cos\frac{n\pi}{2} - 4^n.4^2$

apply z-transform on b.s

$$= Z_T(5n^2) + 4Z_T\left(\cos n\frac{\pi}{2}\right) - Z_T(4^n.16)$$

$$= 5Z_T(n^2) + 4Z_T\left(\cos n\frac{\pi}{2}\right) - 16Z_T(4^n)$$

$$= 5\left(\frac{z^2 z}{(z-1)^3}\right) + 4Z_T\left(\cos n\frac{\pi}{2}\right) - 16\left[\frac{z}{z-4}\right]$$

$$\therefore Z_T\cos^n\theta = \frac{z^2 z\cos\theta}{z^2 - 2z\cos\theta + 1}$$

$$= 5\left(\frac{z^2+z}{(z-1)^3}\right) + \left[4\frac{z^2 z\cos\frac{\pi}{2}}{z^2 2z\cos\pi/2 + 1}\right] - \frac{16z}{z-4}$$

$$= 5\left(\frac{z^2+z}{(z-1)^3}\right) + 4(0) - 16\frac{z}{z-4}$$

$$= 5\left(\frac{z^2+z}{(z-1)^3}\right) - 16\left(\frac{z}{z-4}\right)$$

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8. Find the z-transform of $\cosh\left(\frac{n\pi}{2}+\theta\right)$.

$\Rightarrow$ we have $\cosh(n\theta) = \frac{e^{n\theta}+e^{-n\theta}}{2}$

Now, $\cosh\left(\frac{n\pi}{2}+\theta\right) = \frac{e^{\left(\frac{n\pi}{2}+\theta\right)} + e^{-\left(\frac{n\pi}{2}+\theta\right)}}{2}$

Taking z-transform on b.s

$$Z_T\left[\cosh\left(\frac{n\pi}{2}+\theta\right)\right] = Z_T\left[\frac{e^{\left(\frac{n\pi}{2}+\theta\right)} + e^{-\left(\frac{n\pi}{2}+\theta\right)}}{2}\right]$$

$$= \frac{1}{2} Z_T\left[e^{\frac{n\pi}{2}} \cdot e^{\theta} + e^{\frac{-n\pi}{2}} \cdot e^{-\theta}\right]$$

$$= \frac{1}{2}\left[e^{\theta} \cdot Z_T\left[e^{\frac{n\pi}{2}}\right] + e^{-\theta} \cdot Z_T\left[e^{\left(\frac{-n\pi}{2}\right)}\right]\right]$$

$$= \frac{1}{2}\left[e^{\theta} Z_T\left[\left(e^{\frac{\pi}{2}}\right)^n\right] + e^{-\theta} \cdot Z_T\left[\left(e^{-\pi/2}\right)^n\right]\right]$$

$$= \frac{1}{2}\left[e^{\theta} \times \frac{z}{z-e^{\pi/2}} + e^{-\theta}\frac{z}{z-e^{-\pi/2}}\right]$$

$$= \frac{z}{2}\left[\frac{e^{\theta}}{z-e^{\pi/2}} + \frac{e^{-\theta}}{z-e^{-\pi/2}}\right]$$

$$= \frac{z}{2}\left[\frac{e^{\theta}(z-e^{-\pi/2}) + e^{-\theta}(z-e^{\pi/2})}{(z-e^{\pi/2})(z-e^{-\pi/2})}\right]$$

$$= \frac{z}{2}\left[\frac{ze^{\theta} - e^{\theta}e^{-\pi/2} + ze^{-\theta} - e^{-\theta} \cdot e^{\pi/2}}{z^2 - ze^{-\pi/2} - ze^{\pi/2} + e^{\pi/2} \cdot e^{-\pi/2}}\right]$$

$$= \frac{z}{2}\left[\frac{z(e^{\theta}+e^{-\theta}) - e^{(\theta-\pi/2)} - e^{(-\theta+\pi/2)}}{z^2 - z\left[e^{\pi/2}+e^{-\pi/2}\right]+1}\right]$$

$$= \frac{z}{2}\left[\frac{z(2\cosh\theta) - \left[e^{(\theta-\pi/2)} + e^{-(\theta-\pi/2)}\right]}{z^2 - z\,2\cosh(\pi/2)+1}\right]$$

$$= \frac{z}{2}\left[\frac{2z\cosh\theta - 2\cosh(\theta-\pi/2)}{z^2 - 2z\cosh(\pi/2)+1}\right]$$

$$= \frac{z}{2} \times 2\left[\frac{z\cosh\theta - \cosh(\theta-\pi/2)}{z^2 - 2z\cosh(\pi/2)+1}\right]$$

$$= \frac{z^2\cosh\theta - z\cosh(\theta-\pi/2)}{z^2 - 2z\cosh(\pi/2)+1}$$

9. Find Z-transform of $3^n \cos\left(\frac{n\pi}{4}\right)$

⇒ we have.

$$Z_T[\cos(n\theta)] = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1}$$

$$Z_T\left[\cos n\left(\frac{\pi}{4}\right)\right] = \frac{z^2 - z\cos(\pi/4)}{z^2 - 2z\cos(\pi/4)+1}$$

$$= \frac{z^2 - z\frac{1}{\sqrt{2}}}{z^2 - 2z\frac{1}{\sqrt{2}}+1} = \frac{\frac{\sqrt{2}z^2 - z}{\cancel{\sqrt{2}}}}{\frac{\sqrt{2}z^2 - 2z + \sqrt{2}}{\cancel{\sqrt{2}}}}$$

$$Z_T\left[\cos\left(\frac{n\pi}{4}\right)\right] = \frac{\sqrt{2}z^2 - z}{\sqrt{2}z^2 - 2z + \sqrt{2}} = F[z]$$

$$Z_T(k^n u_n) = F\left(\frac{z}{k}\right)$$

Now,

$$Z_T\left[3^n \cos\left(\frac{n\pi}{4}\right)\right] = F\left(\frac{z}{3}\right)$$

$$= \frac{\sqrt{2}\left(\frac{z}{3}\right)^2 - \frac{z}{3}}{\sqrt{2}\left(\frac{z}{3}\right)^2 - 2\left(\frac{z}{3}\right) + \sqrt{2}}$$

$$= \frac{\sqrt{2}\frac{z^2}{9} - \frac{z}{3}}{\sqrt{2}\frac{z^2}{9} - 2\frac{z}{3} + \sqrt{2}} = \frac{\frac{\sqrt{2}z^2 - 3z}{\cancel{9}}}{\frac{\sqrt{2}z^2 - 6z + 9\sqrt{2}}{\cancel{9}}}$$

$$= \frac{\sqrt{2}z^2 - 3z}{\sqrt{2}z^2 - 6z + 9\sqrt{2}}.$$

INVERSE Z- TRANSFORM

Suppose $F(z)$ be a z-transform $Z_T(U_n)$ then the inverse z-trans ~~$Z_T^{-1}(U_n)$ can~~ $Z_T^{-1}[F(z)]$ can be defined as

$$Z_T^{-1}[F(z)] = U_n$$

1. $Z_T^{-1}\left(\frac{z}{z-k}\right) = k^n$

2. $Z_T^{-1}\left(\frac{z}{z-1}\right) = 1$

3. $Z_T^{-1}\left[\frac{z}{(z-1)^2}\right] = n$

4. $Z_T^{-1}\left[\frac{z^2+z}{(z-1)^3}\right] = n^2$

5. $Z_T^{-1}\left[\frac{z^3+4z^2+z}{(z-1)^4}\right] = n^3$

6. $Z_T^{-1}\left[\frac{z}{z+1}\right] = (-1)^n$

7. $Z_T^{-1}\left[\frac{z}{z^2+1}\right] = \sin\frac{n\pi}{2}$

8. $Z_T^{-1}\left[\frac{z^2}{z^2+1}\right] = \cos n\frac{\pi}{2}$

9. $Z_T^{-1}\left[F\left(\frac{z}{k}\right)\right] = k^n \cdot U_n$

10. $Z_T^{-1}[F(kz)] = k^{-n} \cdot U_n$

1. Find the inverse Z-transform of $\frac{3z^2+2z}{(5z-1)(5z+2)}$

$\Rightarrow$ Given:- $F(z) = \frac{3z^2+2z}{(5z-1)(5z+2)}$

$$F(z) = \frac{z(3z+2)}{(5z-1)(5z+2)}$$

$$\frac{F(z)}{z} = \frac{3z+2}{(5z-1)(5z+2)} = \frac{A}{(5z-1)} + \frac{B}{(5z+2)} \rightarrow ①$$

Now,

$$\frac{3z+2}{(5z-1)(5z+2)} = \frac{A}{(5z-1)} + \frac{B}{(5z+2)}$$

$$\frac{3z+2}{\cancel{(5z-1)(5z+2)}} = \frac{A(5z+2)+B(5z-1)}{\cancel{(5z-1)(5z+2)}} \Rightarrow 3z+2 = A(5z+2)+B(5z-1) \rightarrow ②$$

put $z = -2/5$ in eq^n ②

$$3\left(\frac{-2}{5}\right)+2 = A\left[5\left(\frac{-2}{5}\right)+2\right] + B\left[5\left(\frac{-2}{5}\right)\right] \Rightarrow \frac{-6}{5}+2 = A(0)+B(-2-1)$$

$$\frac{-6+10}{5} = B(-3)$$

$$\frac{4}{5} = -3B \Rightarrow B = -4/15$$

$$\text{eq}^n ① \Rightarrow \frac{F(z)}{z} = \frac{13}{15}\left[\frac{1}{(5z-1)}\right] - \frac{4}{15}\left[\frac{1}{5z+2}\right]$$

$$F(z) = \frac{13}{15}\left[\frac{z}{5z-1}\right] - \frac{4}{15}\left[\frac{z}{5z+2}\right]$$

put $z = 1/5$ in eq^n ①

$$\text{eq}^n ② \Rightarrow 3\left(\frac{1}{5}+2\right) = A\left(\cancel{5}\frac{1}{\cancel{5}}+2\right) + B\left(\cancel{5}\frac{1}{\cancel{5}}-1\right) \Rightarrow \frac{3}{5}+2 = A(1+2) + B(0)$$

$$\frac{13}{5} = 3A$$

$$\boxed{A = 13/15}$$

Apply inverse Z-transform on b.s

$$Z_T^{-1}[F(z)] = Z_T^{-1}\left[\frac{13}{15}\left(\frac{z}{5z-1}\right) - \frac{4}{15}\left(\frac{z}{5z+2}\right)\right] \qquad Z_T^{-1}\left(\frac{z}{z-k}\right) = k^n$$

$$= \frac{13}{15} Z_T^{-1}\left[\frac{z}{5z-1}\right] - \frac{4}{15} Z_T^{-1}\left[\frac{z}{5z+2}\right]$$

$$= \frac{13}{15\times5} Z_T^{-1}\left[\frac{z}{z-1/5}\right] - \frac{4}{15\times5}\left[\frac{z}{z-(-2/5)}\right]$$

$$U_n = \frac{13}{75}\left(\frac{1}{5}\right)^n - \frac{4}{75}\left(\frac{-2}{5}\right)^n$$

02. Find the inverse transform of $\frac{z}{z^2+7z+10}$

$\Rightarrow$ Given $F(z) = \frac{z}{z^2+7z+10}$

$$F(z) = \frac{z}{(z+2)(z+5)} = \frac{z}{(z+2)(z+5)}$$

10
2 5
7

$$\frac{F(z)}{z} = \frac{1}{(z+2)(z+5)} = \frac{A}{(z+2)} + \frac{B}{(z+5)} \rightarrow (1)$$

$$\frac{1}{(z+2)(z+5)} = \frac{A}{z+2} + \frac{B}{z+5}$$

$$1 = A(z+5) + B(z+2) \rightarrow (2)$$

put $z = -5$ in eq (2)

$$1 = A(-5+5) + B(-5+2)$$
$$1 = A(0) + B(-3)$$
$$1 = A(0) - 3B$$
$$1 = -3B$$
$$B = -1/3$$

put $z = -2$ in eq (2)

$$1 = A(-2+5) + B(-2+2)$$
$$1 = A(3) + B(0)$$
$$A = 1/3$$

[Apply inverse z-transform on b.s]^{x}

$$\text{eq}^n (1) \Rightarrow \frac{F(z)}{z} = \frac{1}{3}\left[\frac{1}{z+2}\right] - \frac{1}{3}\left[\frac{1}{z+5}\right]$$

$$= \frac{1}{3}\left[\frac{z}{z+2}\right] - \frac{1}{3}\left[\frac{z}{z+5}\right]$$

Apply inverse - z transform on b.s

$$Z_T^{-1}[F(z)] = Z_T^{-1}\left[\frac{1}{3}\left[\frac{z}{z+2}\right] - \frac{1}{3}\left[\frac{z}{z+5}\right]\right]$$

$$= \frac{1}{3} Z_T^{-1}\left[\frac{z}{z+2}\right] - \frac{1}{3} Z_T^{-1}\left[\frac{z}{z+5}\right]$$

$$= \frac{1}{3} Z_T^{-1}\left[\frac{z}{z-(-2)}\right] - \frac{1}{3} Z_T^{-1}\left[\frac{z}{z-(-5)}\right] \quad \because Z_T^{-1}\left(\frac{z}{z-k}\right) = k^n$$

$$U_n = \frac{1}{3}(-2)^n - \frac{1}{3}(-5)^n$$

---

03. Find the inverse z-transform of $\frac{2z^2 - 3z}{(z+2)(z-4)}$

$A = 1/6 \quad B = 11/6$

$$\Rightarrow F(z) = \frac{z[2z+3]}{(z+2)(z-4)} = \frac{F(z)}{z} = \frac{2z+3}{(z+2)(z-4)}$$

Now,

$$\frac{F(z)}{z} = \frac{2z+3}{(z+2)(z-4)} = \frac{A}{z+2} + \frac{B}{z-4} \rightarrow (1)$$

$$\frac{2Z+3}{(Z+2)(Z-4)} = \frac{A(Z-4)+B(Z+2)}{(Z+2)(Z-4)}$$

$$2Z+3 = A(Z-4)+B(Z+2) \rightarrow ②$$

put $Z=4$ eqⁿ ② put $Z=-2$ in eqⁿ ②

$2(4)+3 = A(4-4)+B(4+2)$

$11 = A(0)+B(6)$

$\boxed{B = 11/6}$

$2(-2)+3 = A(-2-4)+B(0)$

$-1 = -6(A)$

$\boxed{A = 1/6}$

$$\therefore \text{eq}^n ① \Rightarrow \frac{F(Z)}{Z} = \frac{1}{6}\left[\frac{1}{Z+2}\right] + \frac{11}{6}\left[\frac{1}{Z-4}\right]$$

$$F(z) = \frac{1}{6}\left[\frac{Z}{Z+2}\right] + \frac{11}{6}\left[\frac{Z}{Z-4}\right]$$

Apply inverse Z-transform on b.s

$$Z_T^{-1}[F(Z)] = Z_T^{-1}\left[\frac{1}{6}\left(\frac{Z}{Z+2}\right) + \frac{11}{6}\left(\frac{Z}{Z-4}\right)\right]$$

$$= \frac{1}{6} Z_T^{-1}\left[\frac{Z}{Z-(-2)}\right] + \frac{11}{6} Z_T^{-1}\left[\frac{Z}{Z-4}\right]$$

$$\boxed{U_n = \frac{1}{6}(-2)^n + \frac{11}{6}(4^n)}$$

---

04. Find the inverse Z_T of $\dfrac{4Z^2 - 2Z}{Z^3 - 5Z^2 + 8Z - 4}$

$$\Rightarrow F(Z) = \frac{4Z^2-2Z}{Z^3-5Z^2+8Z-4}$$

1	1	−5	8	−4
	0	1	−4	4
	1	−4	4	$\boxed{0}$

$$F(Z) = \frac{Z(4Z-2)}{(Z-1)(Z-2)^2}$$

$$F(Z) = \frac{4Z-2}{(Z-1)(Z-2)}$$

$Z^2 - 4Z + 4 = 0$

$-4 \to -2, -2$

$Z^2 - 2Z - 2Z + 4 = 0$

$Z(Z-2) - 2(Z-2) = 0$

$(Z-2)(Z-2) = 0$

Now, $\frac{4Z-2}{(Z-1)(Z-1)^2} = \frac{A}{(Z-1)} + \frac{B}{(Z-2)} + \frac{C}{(Z-2)^2}$

$$\frac{4Z-2}{(Z-1)(Z-2)^2} = \frac{A(Z-2)^2 + B(Z-1)(Z-2) + C(Z-1)}{(Z-1)+(Z-2)^2}$$

$$4Z-2 = A(Z-2)^2 + B(Z-1)(Z-2) + C(Z-1) \rightarrow (2)$$

put $Z=2$ in eq^n (2)

$$4(2)-2 = A(2-2)^2 + B(2-1)(2-2) + C(2-1)$$

$$8-2 = A(0) + B(0) + C(1)$$

$$\boxed{C=6}$$

put $Z=1$ in eq^n (1)

$$4(1)-2 = A(1-2)^2 + B(1-1)(1-2) + C(1-1)$$

$$4-2 = A(-1)^2 + B(0) + C(0)$$

$$\boxed{2=A}$$

put $Z=0$ in eq^n (2)

$$4(0)-2 = A(0-2)^2 + B(0-1)(0-2) + C(0-1)$$

$$0-2 = 2(4) + 2B - 6$$

$$-2 = 2 + 2B$$

$$-4 = 2B \Rightarrow \boxed{B=-2}$$

eq^n (1) $\Rightarrow \frac{F(Z)}{Z} = \frac{2}{Z-1} - \frac{2}{(Z-2)} + \frac{6}{(Z-2)^2}$

$$F(Z) = 2\left(\frac{Z}{Z-1}\right) - 2\left(\frac{Z}{Z-2}\right) + 6\left[\frac{Z}{(Z-2)^2}\right]$$

Apply inverse Z-transform on b.s.

$$Z_T^{-1}[F(Z)] = Z_T^{-1} \cdot 2\left[\frac{Z}{Z-1}\right] - 2\left[\frac{Z}{Z-2}\right] + 6\left[\frac{Z}{(Z-2)^2}\right]$$

$$2Z_T^{-1}\left[\frac{Z}{Z-1}\right] - 2Z_T^{-1}\left[\frac{Z}{Z-2}\right] + 6Z_T^{-1}\left[\frac{Z}{(Z-2)^2}\right]$$

$$U_n = 2(-1) - Z2^n + 3Z^{-1}\left[\frac{2Z}{(Z-2)^2}\right]$$

$$= 2 - 2^{n+1} + 32^n \cdot n$$

$$U_n = 2 - 2^{n+1} + 3n \cdot 2^n$$

5. Find the inverse Z-transform of $\frac{z^3-20z}{(z-2)^2(z-4)}$

$\Rightarrow$ $F(z) = \frac{z^3-20z}{(z-2)^3(z-4)} = \frac{z(z^2-20)}{(z-2)^3(z-4)} = \frac{z^2-20}{(z-2)^3(z-4)}$ → ①

$$\frac{z^2-20}{(z-2)^3(z-4)} = \frac{A}{z-2} + \frac{B}{(z-2)^2} + \frac{C}{(z-2)^3} + \frac{D}{(z-4)}$$

$z^2-20 = A(z-2)^2(z-4) + B(z-2)(z-4) + C(z-4) + D(z-2)^3$ → ②

put $z=2$ in eq ②

$(2)^2-20 = A(0)+B(0)+C(2-4)+D(2-2)^2$

$4-20 = 0+0-2C+D(0)$

$-16 = -2C$

$C = 8$

put $z=4$ in eq^n ②

$4^2-20 = A(4-2)^2(0)+B(0)+C(0)+D(4-2)^3$

$-4 = D(8)$

$D = -1/2$ ✗

WKT, $Z_T^{-1}\left(\frac{z}{z-2}\right) = 2^n$, $Z_T^{-1}\left[\frac{2z}{(z-2)^2}\right] = 2^n n$

$Z_T^{-1}\left[\frac{2z^2+4z}{(z-2)^3}\right] = 2^n n^2$, $Z_T^{-1}\left(\frac{z}{z-4}\right) = 4^n$

Now, $\frac{z^2-20}{(z-2)^3(z-4)} = \frac{A}{z-2} + \frac{B}{(z-2)^2} + \frac{C(2z+4)}{(z-2)^3} + \frac{D}{z-4}$

$z^2-20 = A(z-2)^2(z-4) + B(z-2)(z-4) + C(2z+4)(z-4) + D(z-2)^3$ → ②

put $z=2$ in eq^n ②

$2^2-20 = A(0)+B(0)+C(4+4)(2-4)+D(0)$

$4-20 = 0+0+C(8)(-2)+0$

$-16 = -16C$

$\boxed{C=1}$

put $z=4$ in eq^n ②

$4^2-20 = A(0)+B(0)+C(0)+D(4-2)^3$

$\boxed{D=-1/2}$

on comparing z^3 co-efficient on b.s

$0 = A + D$

$A = -D$

$= -(-1/2) = \frac{1}{2}$

put $z = 0$ in eq^n (2)

$0^2 - 20 = A(-2)^2(-4) + B(-2)(-4) + C(4)(-4) + D(-2)^3$

$-20 = A(4)(-4) + 8B - 16C - 8D$

$-20 = \frac{1}{2}(-16) + 8B - 16(1) - 8(-1/2)$

$-20 = -8 + 8B - 16 + 4$

$-20 = -20 + 8B$

$-20 + 20 = 8B$

$0 = 8B$

$\boxed{B = 0}$

On substituting A, B, C, D in eq^n (1)

$$\frac{F(z)}{z} = \frac{1}{2}\left[\frac{1}{(z-2)}\right] + 0 + \left[\frac{2z+4}{(z-2)^3}\right] - \frac{1}{2}\left[\frac{1}{z-4}\right]$$

$$F[z] = \frac{1}{2}\left[\frac{z}{z-2}\right] + \left[\frac{z(2z+4)}{(z-2)^3}\right] - \frac{1}{2}\left[\frac{z}{z-4}\right]$$

Take inverse z-transform on b.s.

$$Z_T^{-1}[F(z)] = \frac{1}{2} Z_T^{-1}\left(\frac{z}{z-2}\right) + Z_T^{-1}\left[\frac{2z^2+4z}{(z-2)^3}\right] - \frac{1}{2} Z_T^{-1}\left[\frac{z}{z-4}\right]$$

$= \frac{1}{2} 2^n + 2^n . n^2 - \frac{1}{2} 4^n$

$= 2^n . 2^{-1} + 2^n . n^2 - (2^2)^n . 2^{-1}$

$= 2^{n-1} + 2^n . n^2 - 2^{2n-1}$

Difference equation

A difference equation is a relationship between differences of an unknown function [dependent variable] at several values of independent variable.

Ex:- $U_{n+2} - 4U_n = 0$

$y_{n+2} + 6y_{n+1} + 9y_n = 2n$

Working rule

1. * Take Z-Transform on b.s of given D.E
   * An use $Z_T(U_n) = \bar{U}(z)$

$$Z_T(U_{n+1}) = z(\bar{U}(z) - U_0)$$

$$Z_T(U_{n+2}) = z^2\left(\bar{U}(z) - U_0 - \frac{U_1}{z}\right)$$

2. write $\bar{U}(z)$ as a function of z hence applied inverse z transform and find U_n.

PROBLEMS

1. Solve difference eq^n using Z-transform $U_{n+2} - 4U_n = 0$, subject to the conditions $U_0 = 0$, $U_1 = 2$.

$\Rightarrow$ Given:

$U_{n+2} - 4U_n = 0$, $U_0 = 0$, $U_1 = 2$

Apply Z-transform on b.s

$$Z_T(U_{n+2} - 4U_n) = Z_T(0)$$

$$Z_T(U_{n+2}) - 4\,Z_T(U_n) = 0$$

$$z^2\left(\bar{U}(z) - U_0 - \frac{U_1}{z}\right) - 4\bar{U}(z) = 0$$

$$z^2\left(\bar{U}(z) - 0 - \frac{2}{z}\right) - 4\bar{U}(z) = 0$$

$$z^2\left(\bar{U}(z) - \frac{2}{z}\right) - 4\bar{U}(z) = 0$$

$$z^2\bar{U}(z) - \frac{2}{z}(z^2) - 4\bar{U}(z) = 0$$

$$z^2\bar{U}(z) - 2z - 4\bar{U}(z) = 0$$

$$(z^2 - 4)\bar{U}(z) - 2z = 0$$

$$(z^2 - 4)\bar{U}(z) = 2z$$

$$\bar{U}(z) = \frac{2z}{z^2-4}$$

$$\frac{\bar{U}(z)}{z} = \frac{2}{z^2-4} = \frac{2}{z^2-2^2}$$

$$\frac{\bar{U}(z)}{z} = \frac{2}{(z-2)(z+2)} = \frac{A}{z-2} + \frac{B}{z+2} \rightarrow ①$$

$$2 = A(z+2) + B(z-2) \rightarrow ②$$

put $z=2$ eq^n ②

$$2 = A(2+2) + B(2-2)$$

$$2 = A(4) + B(0)$$

$$\boxed{A = 1/2}$$

put $z=-2$ in eq^n ②

$$2 = A(-2+2) + B(-2-2)$$

$$2 = A(0) + B(-4)$$

$$\frac{2}{-4} = B \Rightarrow \boxed{B = -1/2}$$

substitute A & B in eq^n ①

$$\frac{\bar{U}(z)}{z} = \frac{1/2}{(z-2)} + \frac{-1/2}{(z+2)} \Rightarrow \frac{1}{2}\left(\frac{1}{z-2}\right) - \frac{1}{2}\left(\frac{1}{z+2}\right)$$

$$\bar{U}(z) = \frac{1}{2}\left[\frac{z}{z-2}\right] - \frac{1}{2}\left[\frac{z}{z-(-2)}\right]$$

apply inverse z-transform of b.s

$$Z_T^{-1}[\bar{U}(z)] = \frac{1}{2} Z_T^{-1}\left[\frac{z}{z-2}\right] - \frac{1}{2} Z_T^{-1}\left[\frac{z}{z-(-2)}\right]$$

$$= \frac{1}{2}(2^n) - \frac{1}{2}(-2)^n$$

---

02. Solve the difference eq^n using z-transform $U_{n+2} + 4U_{n+1} + 3U_n = 3^n$ subject to the conditions $U_0 = 0 \quad U_1 = 1$

$\Rightarrow$

$$U_{n+2} + 4U_{n+1} + 3U_n = 3^n \qquad U_0 = 0 \quad U_1 = 1$$

apply z-transform on b.s

$$Z_T(U_{n+2} + 4U_{n+1} + 3U_n) = Z_T(3^n)$$

$$Z_T(U_{n+2}) + Z_T(4U_{n+1}) + Z_T(3U_n) = Z_T(3^n)$$

$$z^2\left[\bar{U}(z) - U_0 - \frac{U_1}{z}\right] + 4z\left[\bar{U}(z) - U_0\right] + 3\left[\bar{U}(z)\right] = \frac{z}{z-3}$$

$$z^2\left[\bar{U}(z) - 0 - \frac{1}{z}\right] + 4z\left[\bar{U}(z) - 0\right] + 3\left[\bar{U}(z)\right] = \frac{z}{z-3}$$

$$z^2\left[\bar{U}(z)-\frac{1}{z}\right]-4z\left[\bar{U}(z)\right]+3\bar{U}(z)=\frac{z}{z-3}$$

$$z^2\bar{U}(z)-\frac{z^2}{z}+4z\bar{U}(z)+3\bar{U}(z)=\frac{z}{z-3}$$

$$\bar{U}(z)(z^2+4z+3)-z=\frac{z}{z-3}$$

$$(z+3)(z+1)\,\bar{U}(z)=\frac{z}{z-3}+z$$

$$(z+3)(z+1)\,\bar{U}(z)=\frac{z+z(z-3)}{z-3}$$

$$\bar{U}(z)=\frac{z+z(z-3)}{(z-3)(z+1)(z+3)} \Rightarrow \frac{z^2-2z}{(z-3)(z+1)(z+3)}$$

$$\bar{U}(z)=\frac{z(z-2)}{(z-3)(z+1)(z+3)} \Rightarrow \frac{\bar{U}(z)}{z}=\frac{z-2}{(z-3)(z+1)(z+3)}$$

$$\frac{\bar{U}(z)}{z}=\frac{A}{z-3}+\frac{B}{z+1}+\frac{C}{z+3} \quad \text{--- (1)}$$

$$z-2=A(z+1)(z+3)+B(z-3)(z+3)+C(z-3)(z+1) \quad \text{--- (2)}$$

put $z=3$ in eqn (2)

$$3-2=A(3+1)(3+3)+B(3-3)(3+3)+C(3-3)(3+1)$$

$$1=A(24)+B(0)+C(0)$$

$$A=1/24$$

put $z=-3$ in eqn (2)

$$-3-2=A(-3+1)(-3+3)+B(-3-3)(-3+3)+C(-3-3)(-3+1)$$

$$-5=A(0)+B(0)+C(-6)(-2)$$

$$-5=12C$$

$$\boxed{C=-5/12}$$

put $z=-1$ in eqn (2)

$$-1-2=A(-1+1)(-1+3)+B(-1-3)(-1+3)+C(-1-3)(-1+1)$$

$$-3=A(0)+B(-4)(2)+C(0)$$

$$-3=B(-8)$$

$$\boxed{B=3/8}$$

substitute A, B, C in eqⁿ 1

$$\frac{\bar{U}(z)}{z} = \frac{A}{z-3} + \frac{B}{z+1} + \frac{C}{z+3}$$

$$= \frac{1}{24}\left[\frac{1}{z-3}\right] + \frac{3}{8}\left[\frac{1}{z+1}\right] - \frac{5}{12}\left[\frac{1}{z+3}\right]$$

$$\bar{U}(z) = \frac{1}{24}\left[\frac{z}{z-3}\right] + \frac{3}{8}\left[\frac{z}{z-(-1)}\right] - \frac{5}{12}\left[\frac{z}{z-(-3)}\right]$$

apply inverse z-transform on b.s

$$Z_T^{-1}\left[\bar{U}(z)\right] = \frac{1}{24} Z_T^{-1}\left[\frac{z}{z-3}\right] + \frac{3}{8} Z_T^{-1}\left[\frac{z}{z-(-1)}\right] - \frac{5}{12} Z_T^{-1}\left[\frac{z}{z-(-3)}\right]$$

$$= \frac{1}{24}(3^n) + \frac{3}{8}(-1)^n - \frac{5}{12}(-3)^n$$

Imp

03. Solve the difference $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using z-transforms.

⇒ Given :- $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$, $y_0 = y_1 = 0$

apply z-transform on b.s

$$Z_T(y_{n+2} + 6y_{n+1} + 9y_n) = Z_T(2^n)$$

$$Z_T y_{n+2} + 6 Z_T y_{n+1} + 9 Z_T y_n = Z_T 2^n$$

$$z^2\left[\bar{y}(z) - y_0 - \frac{y_1}{z}\right] + 6z\left[\bar{y}(z) - y_0\right] + 9\bar{y}(z) = \frac{z}{z-2}$$

$$z^2\left[\bar{y}(z) - 0 - 0\right] + 6z\left[\bar{y}z - 0\right] + 9\bar{y}(z) = \frac{z}{z-2}$$

$$z^2\bar{y}(z) + 6z\bar{y}(z) + 9\bar{y}(z) = \frac{z}{z-2}$$

$$(z^2 + 6z + 9)\bar{y}(z) = \frac{z}{z-2}$$

$$(z+3)(z+3)\bar{y}(z) = \frac{z}{z-2}$$

$$\bar{y}(z) = \frac{z}{(z-2)(z+3)(z+3)} = \frac{z}{(z-2)(z+3)^2}$$

$$\frac{\bar{y}(z)}{z} = \frac{1}{(z-2)(z+3)^2} = \frac{A}{z-2} + \frac{B}{(z+3)^2} + \frac{C}{(z+3)^2} \quad \text{—(1)}$$

$\cancel{1 = A(z+3)^2 + B(z-2) \text{ —(2)}}$

~~put z = 2 in eqⁿ 1~~

$\cancel{1 = A(2+3)^2 + B(2-2)}$

$\cancel{1 = 25A + B(0)}$

$\cancel{A = 1/25}$

~~Put z=~~ $\dfrac{1}{(z-2)(z+3)^2} = \dfrac{A}{z-2} + \dfrac{B}{z+3} + \dfrac{C}{(z+3)^2}$

$1 = A(z+3)^2 + B(z-2)(z+3) + C(z-2) \rightarrow (2)$

put $z = -3$ in eq^n (2)

$1 = A(-3+3)^2 + B(-3-2)(-3+3) + C(-3-2)$

$1 = A(0) + B(0) + C(-5)$

$\boxed{-1/5 = C}$

put $z = 2$ in eq^n (2)

$1 = A(2+3)^2 + B(2-2)(2+3) + C(2-2)$

$1 = 25A + B(0) + C(0)$

$\boxed{1/25 = A}$

put $z = 0$ in eq (2)

$1 = A(0+3)^2 + B(0-2)(0+3) + C(0-2)$

$1 = 9A + (-6)B - 2C$

$1 = 9\left(\frac{1}{25}\right) - 6B + 2\left(\frac{1}{5}\right)$

$1 = \frac{9}{25} - 6B + \frac{2}{5}$

$1 = \frac{9}{25} + \frac{2}{5} - 6B \Rightarrow \frac{9+10}{25} - 6B$

$1 - \frac{19}{25} = -6B$

$\frac{25-19}{25} = -6B$

$\frac{\cancel{6}}{25} = -\cancel{6}B$

$\boxed{B = -1/25}$

substitute A, B, C in eq^n (1)

$\dfrac{Y(z)}{z} = \dfrac{1}{(z-2)(z+3)^2} = \dfrac{A}{z-2} + \dfrac{B}{z+3} + \dfrac{C}{(z+3)^2}$

$\dfrac{Y(z)}{z} = \dfrac{1/25}{z-2} + \dfrac{(-1/25)}{z+3} - \dfrac{1/5}{(z+3)^2}$

$$\frac{\bar{Y}(z)}{z} = \frac{1}{25}\left[\frac{1}{z-2}\right] - \frac{1}{25}\left[\frac{1}{z+3}\right] - \frac{1}{5}\left[\frac{1}{(z+3)^2}\right]$$

$$\bar{Y}(z) = \frac{1}{25}\left[\frac{z}{z-2}\right] - \frac{1}{25}\left[\frac{z}{z+3}\right] - \frac{1}{5}\left[\frac{z}{(z+3)^2}\right]$$

Take inverse transform on b.s

$$Z_T^{-1}\left[\bar{Y}(z)\right] = \frac{1}{25} Z_T^{-1}\left[\frac{z}{z-2}\right] - \frac{1}{25} Z_T^{-1}\left[\frac{z}{z+3}\right] - \frac{1}{5} Z_T^{-1}\left[\frac{z}{(z+3)^2}\right]$$

$$= \frac{1}{25}2^n - \frac{1}{25}(-3)^n - \frac{1}{5\times -3} Z_T^{-1}\left[\frac{-3z}{(z-(-3))^2}\right]$$

$$= \frac{1}{25}2^n - \frac{1}{25}(-3)^n + \frac{1}{15}(-3^n)\cdot n$$

4. Solve the difference eq^n $u_{n+2} + 2u_{n+1} + u_n = n$, $u_0 = 0$, $u_1 = 0$ by using z-transform.

$\Rightarrow$ Given: $u_{n+2} + 2u_{n+1} + u_n = n$, $u_0 = 0$, $u_1 = 0$

Apply z transform on b.s

$$Z_T(u_{n+2} + 2u_{n+1} + u_n) = Z_T(n)$$

$$Z_T u_{n+2} + 2Z_T u_{n+1} + Z_T u_n = Z_T n$$

$$\left(z^2\bar{u}(z) - u_0 - \frac{u_1}{z}\right) + 2\left(z\bar{u}(z) - u_0\right) + \bar{u}(z) = \frac{z}{(z-1)^2}$$

$$\left(z^2\bar{u}(z) - 0 - \frac{0}{z}\right) + 2\left(z\bar{u}(z) - 0\right) + \bar{u}(z) = \frac{z}{(z-1)^2}$$

$$z^2\bar{u}(z) + 2z\bar{u}(z) + \bar{u}(z) = \frac{z}{(z-1)^2}$$

$$\bar{u}(z)\left[z^2 + 2z + 1\right] = \frac{z}{(z-1)^2}$$

$$\bar{u}(z)(z+1)(z+1) = \frac{z}{(z-1)^2}$$

$$\bar{u}(z) = \frac{z}{(z-1)^2(z+1)(z+1)} = \frac{z}{(z-1)^2(z+1)^2}$$

$$\frac{\bar{u}(z)}{z} = \frac{1}{(z-1)^2(z+1)(z+1)} = \frac{1}{(z-1)^2(z+1)^2} = \frac{A}{(z-1)} + \frac{B}{(z-1)^2} + \frac{C}{z+1} + \frac{D}{(z+1)^2} \quad \to (1)$$

$$\frac{1}{(z-1)^2(z+1)^2} = \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{(z+1)} + \frac{D}{(z+1)^2}$$

$$1 = A(z-1)(z+1)^2 + B(z+1)^2 + C(z-1)^2(z+1) + D(z-1)^2 \quad \to (2)$$

put $z=1$

$1 = A(1-1)(1+1)^2 + B(1+1)^2 + C(1-1) + D(1-1)$

$1 = 0 + 4B + C(0) + D(0)$

$\boxed{B = 1/4}$

put $z = -1$

$1 = A(-1-1)(-1+1)^2 + B(-1+1)^2 + C(-1-1)^2(-1+1) + D(-1-1)^2$

$1 = A(0) + B(0) + C(0) + (-2)^2 D$

$\boxed{D = 1/4}$

put $z=0$ in eq (2)

$1 = A(0-1)(0+1)^2 + B(0+1)^2 + C(0-1)^2(0+1) + D(0-1)^2$

$1 = -A + B + C + D$

$1 = -A + \frac{1}{4} + C + \frac{1}{4}$

$= -A + \frac{1}{4} - A + \frac{1}{4}$

$1 = -2A + \frac{1}{2}$

$1 - \frac{1}{2} = -2A$

$\frac{1}{2} = -2A \Rightarrow \boxed{A = -\frac{1}{4}}$

from eq^n (2)

$1 = A(z-1)(z^2+1+2z) + B(z^2+1+2z) + C(z+1)(z^2+1-2z) + D(z^2+1-2$

$= A(z^3+z+2z^2-z^2-1-2z) + Bz^2+B+2Bz + C(z^3+z-2z^2+z^2+1$

$-2z) + Dz^2+D-2Dz$

$= A(z^3+z^2-z-1) + Bz^2 + B + 2Bz + C(z^3-z^2-z+1) +$
$Dz^2+D-2Dz$

$1 = Az^3 + Az^2 - Az - A + Bz^2 + B + 2Bz + Cz^3 \; Cz^2 \; Cz + C + Dz^2 +$
$D - 2Dz$

$0z^3 + 0z^2 + 0z + 1 = (A+C)z^3 + (A+B-C+D)z^2 + (-A+2B-C-2D)z +$
$(-A+B+C+D)$

on comparing z^3 coefficients on b.s

$0 = A + C$

$\boxed{A = -C}$

we have, $A = -C$

$$C = -A$$

$$C = -\left(-\frac{1}{4}\right)$$

$$\boxed{C = 1/4}$$

substitute A B C D in eq^n (1)

$$\frac{\bar{U}(z)}{z} = \frac{A}{(z-1)} + \frac{B}{(z-1)^2} + \frac{C}{z+1} + \frac{D}{(z+1)^2}$$

$$= \frac{-1}{4}\left[\frac{1}{z-1}\right] + \frac{1}{4}\left[\frac{1}{(z-1)^2}\right] + \frac{1}{4}\left[\frac{1}{z+1}\right] + \frac{1}{4}\left[\frac{1}{(z+1)^2}\right]$$

$$\bar{U}(z) = \frac{-1}{4}\left[\frac{z}{z-1}\right] + \frac{1}{4}\left[\frac{z}{(z-1)^2}\right] + \frac{1}{4}\left[\frac{z}{z+1}\right] + \frac{1}{4}\left[\frac{z}{(z+1)^2}\right]$$

apply inverse z - transform on b.s

$$Z_T^{-1}[\bar{U}(z)] = -\frac{1}{4} Z_T^{-1}\left[\frac{z}{z-1}\right] + \frac{1}{4} Z_T^{-1}\left[\frac{z}{(z-1)^2}\right] + \frac{1}{4} Z_T^{-1}\left[\frac{z}{z+1}\right]$$

$$+ Z_T^{-1}\left[\frac{z}{(z+1)^2}\right]$$

$$= -\frac{1}{4}(1) + \frac{1}{4}(n) + \frac{1}{4}(-1)^n - \frac{1}{4}(-1)^n n$$

$$= 1/4\,(-1+n) + [(-1)^n - (-1)^n n]$$

$$= 1/4\,[(n-1) - (-1)^n (n-1)]$$

$$Z_T^{-1}[\bar{U}(z)] = \frac{n-1}{4}\left[1 - (-1)^n\right]$$

H.W $U_{n+2} - 4U_n = n-1$ with $U_0 = 1$, $U_1 = 2$